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PALEOMAGNETIC RESULTS FROM HONDURAS AND THEIR BEARING ON CARIBBEAN TECTONICS

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Abstract. Paleomagnetic results have been obtained from 37 sites in central and western Honduras and one site in northwestern Nicaragua. The rocks are limestones, sandstones, and granites and range in age from 140 to 60 m.y. Regional fold-tests demonstrate that the rocks carry a prefolding (pre-Laramide) magnetization. The data indicate a very large clockwise rotation of the sampling area during the Early Cretaceous followed by a counterclockwise rotation in the Late Cretaceous. The changes in latitude are minor during this time period. It is highly unlikely that these large rotations apply to the entire Caribbean plate. Rather, they reflect the motion of a smaller area known as the Chortis Block. This interpretation is supported by paleomagnetic data from neighboring areas. The results also suggest that the Chortis Block sutured onto southern Mexico/Guatemala in Latest Cretaceous time in agreement with geochemical data and several plate tectonic models. These new data together with published paleomagnetic data from around the Caribbean imply that a rigid Caribbean plate did not exist until well into the Tertiary.

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INTRODUCTION

In one of the earliest papers on the Caribbean region, Vaughan [1918] divided this area into eight "major tectonic provinces," of which four lie in Central America. Dengo [1969] and Dengo and Bohnenberger [1969] proposed a strikingly similar subdivision for Central America except that they did not regard the Yucatan Platform as a separate tectonic unit. Dengo [1969] introduced the names Maya Block, Chortis Block, and southern Central American Block. The Maya Block extends from the Isthmus of Tehuantepec in southern Mexico to the Motagua fault in Guatemala, the Chortis Block from the Motagua fault to central Nicaragua, and the southern Central America Block extends south to the Atrato Lowlands of western Colombia (Figure 1). Because of the lack of geological information, the boundary between the Chortis Block and southern Central America is poorly constrained, and Dengo and Bohnenberger [1969] suggest parallel $12^{o}30'N$ for its approximate location.

The Maya and Chortis Blocks are referred to collectively as northern or nuclear Central America [Schuchert, 1935]. Nuclear Central America has a continental crust, and metamorphic and plutonic rocks of Paleozoic and possibly Late Precambrian age are overlain by sedimentary rocks as old as Late Paleozoic [e.g., Dengo, 1969; Bonis et al., 1970; Weyl, 1980; Gose, 1985]. In southern Central America the

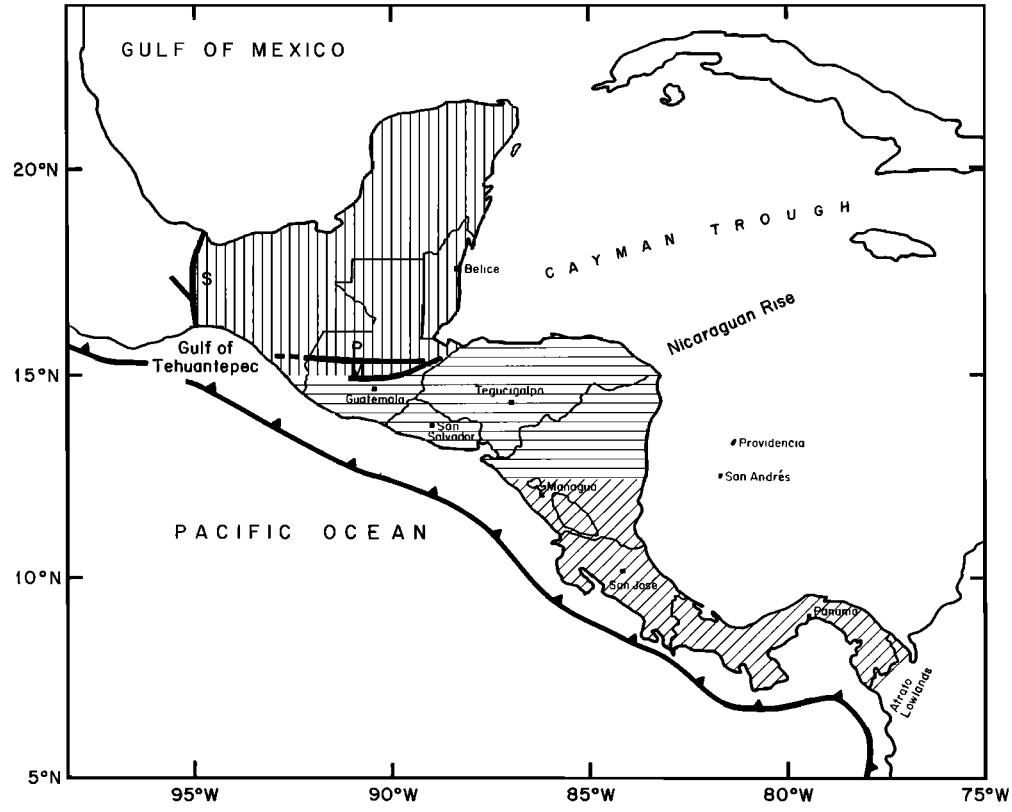


Fig. 1. Reference map of Central America. The Maya Block, Chortis Block, and southern Central American Block as defined by Dengo and Bohnenberger [1969] are indicated by vertical, horizontal and diagonal hatchured patterns. The heavy lines show the Salina Cruz fault (S), Polochic fault (P), Motagua fault (M), and the Middle America trench.

oldest known rocks are ophiolites of Late Jurassic age (Nicoya Complex, Schmidt-Effing [1979]), and the overlying sedimentary strata are Late Cretaceous and younger. Based on gravity data and the occurrence of ophiolitic rocks, it is generally accepted that southern Central America is built on oceanic crust [Dengo, 1969; de Boer, 1974, 1979; Case, 1974; Case and Holcombe, 1980; Weyl, 1980]. In the present plate tectonic setting, the Maya Block is part of the North American plate, whereas the Chortis and southern Central America Blocks lie on the Caribbean plate [Molnar and Sykes, 1969].

This paper reports on paleomagnetic data from Honduras, i.e., the Chortis Block. Gose and Swartz [1977] and Gose et al. [1978] showed this block to have experienced very large rotations throughout the Cretaceous and argued that the Chortis Block was not rigidly connected to the Caribbean plate, at least during that time. This interpretation was challenged by MacDonald [1978, 1980], who pointed out that the rotations may only apply to local, supracrustal blocks or may be apparent rotations due to incorrect tilt corrections. The new data presented here confirm the large rotations and strongly support the validity of the data for plate tectonic interpretations.

GENERAL GEOLOGY AND PROCEDURES

The exposed rocks in Honduras define three major sequences: a poorly mapped metamorphic basement of Paleozoic or older age, a Mesozoic (mainly Cretaceous) sedimentary suite of clastic and carbonaceous rocks, and a volcanic sequence, dominated by ignimbrites, of Tertiary age [Elvir, 1974]. The Mesozoic stratigraphy of central Honduras, the only part of the Republic that has been mapped to any detail, has been described by Finch [1981]. This

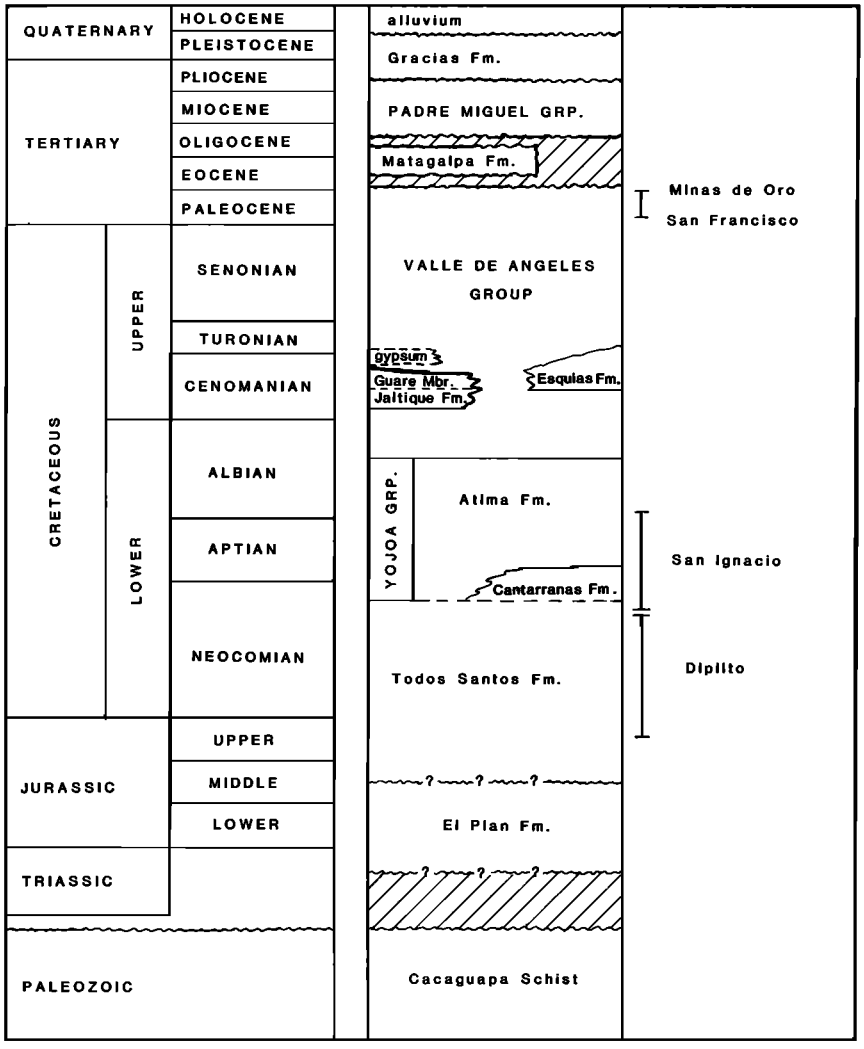


Fig. 2. Stratigraphic column for central Honduras [after Finch, 1981] and age range of four intrusives used in this study.

study focused on the Mesozoic sedimentary rocks and was designed in close cooperation with R. Finch.

The oldest sedimentary unit exposed in Honduras is the El Plan Formation, a black shale of (?) Late Triassic/Jurassic age (Figure 2). It is overlain by the Jurassic/Cretaceous clastic Todos Santos Formation. As will be shown in a separate paper, these strata are not part of the unit known as the Todos Santos Formation of northern Guatemala and southern Mexico. The next higher stratigraphic unit is the Yojoa Group, consisting of a shaly limestone, the Cantarranas Formation (Neocomian to Early Aptian) and a thick, massive-bedded micrite/biomicrite, the Atima Formation (Aptian and Albian). At several locations, the Atima Formation is in gradational contact with the red beds of the overlying Valle de Angeles Group. These red beds are separated into a lower and upper member by two Cenomanian limestones, the Esquias and Jaitique Formations. The upper age limit of the Valle de Angeles Group is poorly constrained. It is unconformably overlain by Middle Tertiary volcanic rocks. Our paleomagnetic data suggest that the group is probably restricted to the Cretaceous. We sampled the entire sequence of Mesozoic sedimentary rocks as well as four dated intrusives ranging in age from 140 to 59 m.y.

In the tropical environment of Hon-

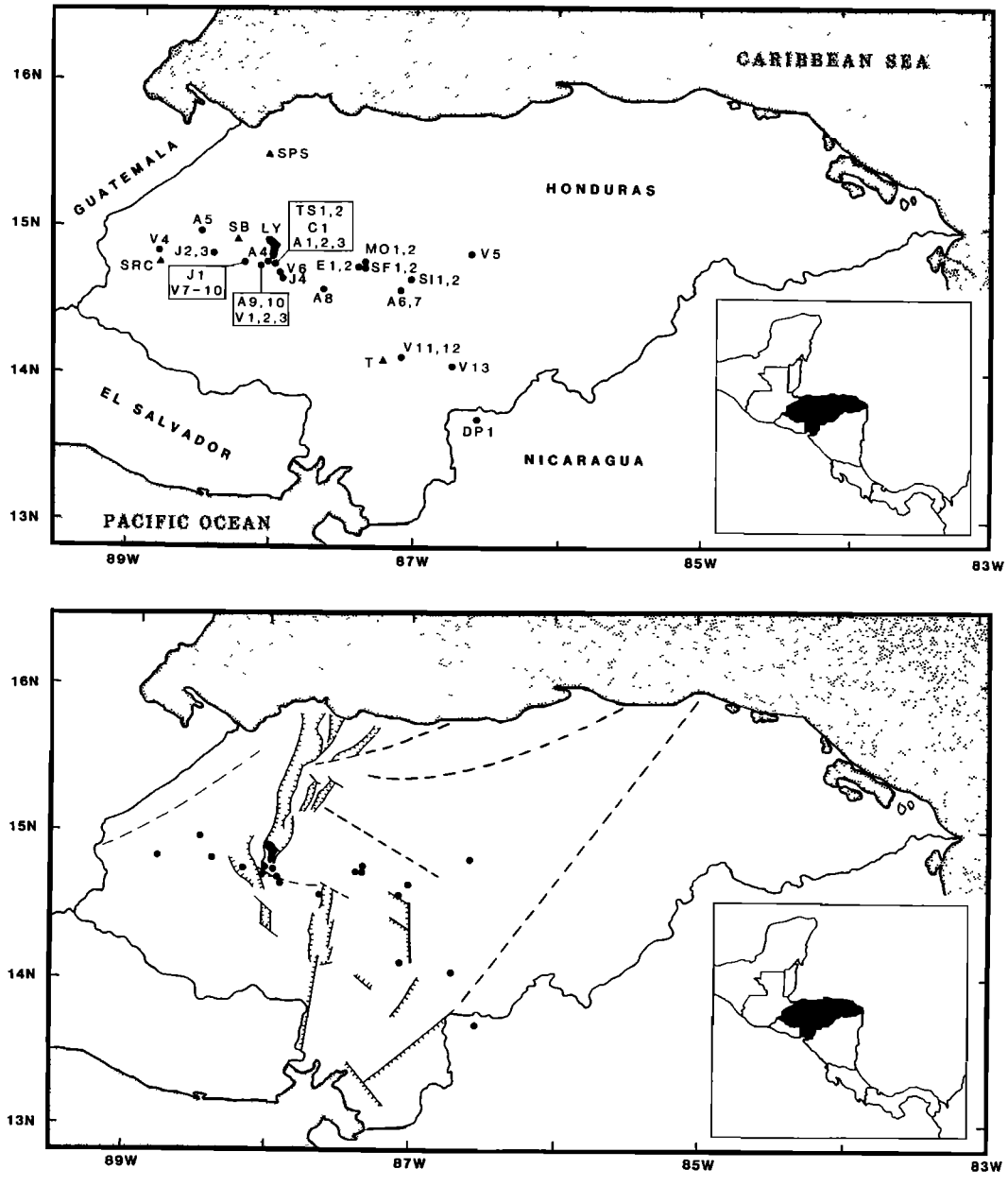


Fig. 3. Distribution of sampling sites (solid circles) referred to in the text. Triangles mark cities: T Tegucigalpa, SB Santa Barbara, SRC Santa Rosa de Copan, SPS San Pedro Saula. LY stands for Lake Yojoa. Bottom figure shows the sites relative to major faults. Solid lines indicate normal faults with barbs on the downthrown side, dashed lines mark faults of unknown displacements (structures after Muehlberger [1976]).

duras, it is difficult to find suitable outcrops for paleomagnetic sampling. In addition to pervasive weathering, slumping is a major problem and the lush vegetation often precludes tracing of beds beyond the immediate outcrop area in order to test for structural continuity. To minimize these problems, most sampling sites were located at recent road cuts or in creek beds where erosion keeps pace with weathering. By collecting at many sites (Figure 3) consistency among the

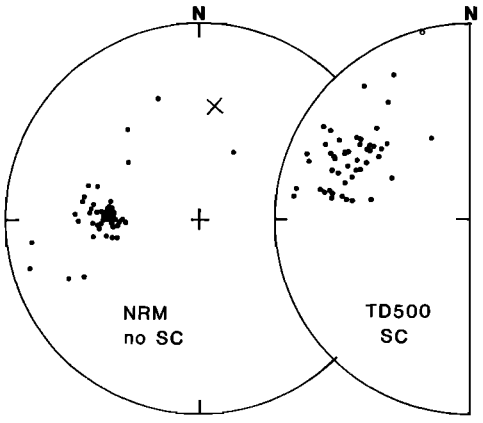


Fig. 4. Directions of magnetization for samples from the Todos Santos Formation (site TS1). The NRM directions are plotted with no structural correction (no SC) and the directions after thermal demagnetization at 500°C (TD500) with structural correction (SC). Solid (open) circles are in the lower (upper) hemisphere of this Schmidt equal area projection. Same convention applies to all equivalent figures. Note that the NRM directions are well removed from the present field direction (cross).

data can be used as a check. Twenty-one out of 59 sampling sites had to be rejected because (1) no internally consistent direction could be obtained, (2) the samples were remagnetized, or (3) the results disagreed with other equivalent data implying slumping.

The results from the first field season (11 sites) have been published by Gose and Swartz [1977]. Because of the subsequent acquisition of a magnetically shielded room, these samples were remeasured and further demagnetized. For the limestones, switching from AF to thermal demagnetization resulted in a significant change of the direction of magnetization, particularly for samples from the Yojoa Group (pole 1 and 2 of Gose and Swartz [1977]). All the initial data have been reanalyzed and are incorporated in this report.

The rocks were stored in a magnetically shielded room (~100 gammas) where they remained throughout the experimental procedure. All samples were progressively demagnetized, typically in five to eight steps. The field in the cooling region as well as around the AF demagnetization coil was less than 3 gammas. The data were analyzed either by applying Fisher statistics to a particular demagnetization step or by calculating linear trends in orthogonal vector projections (As-Zijderveld diagrams). For many sites, the directions of magnetization were extremely stable after the first heating (200°C) and As-Zijderveld diagrams were only used for verification.

EXPERIMENTAL RESULTS

Sedimentary Rocks

Todos Santos Formation (TS1, TS2). The Todos Santos Formation was collected at four sites just south of Lake Yojoa near Pito Solo. All sites are close to the contact with the overlying Yojoa Group. Samples from two sites contain only one component of magnetization which forms a tight cluster well removed from the present field direction (Figure 4). After thermal demagnetization to 550°C, the directions start to disperse. This implies that the characteristic direction of magnetization is carried by magnetite, a strong argument that these rocks retained their primary, i.e., depositional remanent magnetization. No useful data were obtained from the other two sites located in the bed of the nearby Rio Tepemechin.

Cantarranas Formation (C1). At the Pito Solo site, the Cantarranas Formation consists of two distinct lithologies, grey limestone alternating with yellow marlstone. The NRM directions of both rock types are closely aligned with the present field direction for Honduras (Figure 5a). Upon AF demagnetization to 30 mT, most marlstones retain the present field direction, whereas the directions of the limestones move towards the southwest (Figure 5b). This separation becomes more pronounced upon heating (Figure 5c). Figure 5d shows the directions for all samples from which a characteristic magnetization could be extracted. The limestones define a mean direction about 80° removed from the present field, whereas the marlstones form a broad streak between these two directions. The marlstones have been omitted from the statistics.

Atima Formation (A1-A10). The massive limestones of the Atima Formation (up to 1210 m thick) are exposed across much of central Honduras where they commonly form the crests of mountains. Site 1 represents the first 8 m above the contact with the Cantarranas Formation, site 2 spans the interval between 100 and 250 m upsection at the same roadside exposure, and

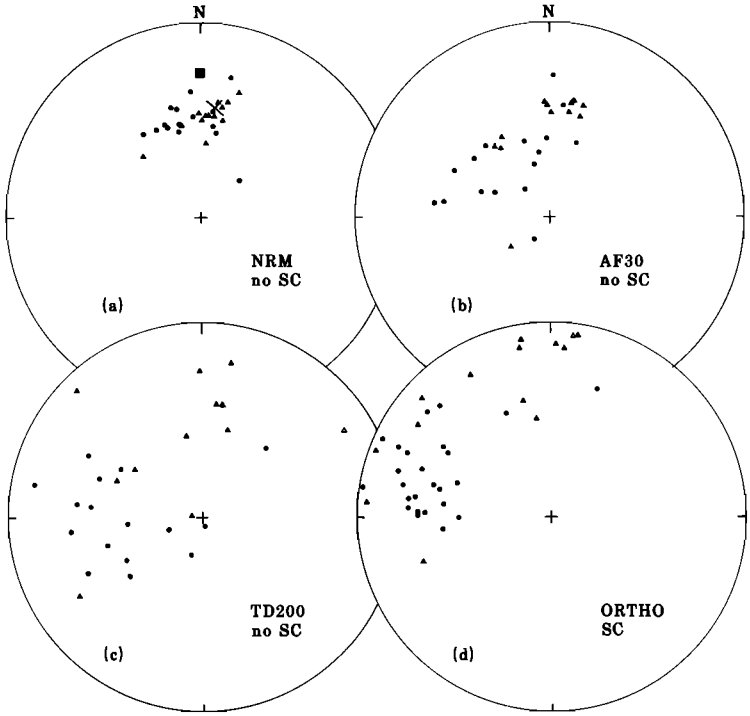


Fig. 5. Cantarranas Formation at Pito Solo (site C1). Circles represent grey limestones and triangles yellow marlstones. The directions of the marlstones are aligned with the present field direction (cross) rather than the theoretical dipole direction (solid square) implying a very recent magnetization. AF stands for alternating field demagnetization at specified value (in millitesla) and ORTHO refers to directions obtained from the analysis of orthogonal vector projections.

site 3 is about 50 m above the contact along the nearby Rio Tepemechin. All samples were progressively demagnetized, and the characteristic direction was usually obtained only upon thermal demagnetization (Figure 6). About half the samples from site 2 became too weakly magnetized after alternating field demagnetization to 30 mT to permit thermal treatment and have been excluded from the statistics.

A measured section near Buena Fe [Finch, 1972], some 7 km west of the previous sites, constitutes site 4. The NRM directions exhibit a large scatter with both polarities, but upon thermal demagnetization the directions form a tight cluster (Figure 7a). At the type section near Atima [Mills et al., 1967], we collected a 10-m stratigraphic interval in a cave approximately 150 to 200 m below the upper contact (site 5). This is the only site from the Atima Formation which is reversely magnetized. AF demagnetization to 20 mT only partially removed a present field overprint, whereas mild heating readily revealed the stable direction in 11 out of 22 samples (Figure 7b). These 11 samples have a mean intensity of 11.1 x 10^{-8} Am^2/kg, while the rejected samples average only 0.67 x 10^{-8} Am^2/kg, a value close to the noise level of the magnetometer.

A steep mountain road near Cedros exposes a structurally complex section of the Atima Formation, including overturned beds [King, 1973]. Two of these sites (sites 6 and 7) yield the same direction (Figure 7c). For the third site, simple rotation around strike is apparently not the proper structural correction. This site is either a slump block or was subjected to multiple deformation, and these results have been discarded.

The upper portion of the Atima Formation was collected near La Libertad in central Honduras (site 8) and along Rio Guajiniquil in western Honduras (sites 9 and 10). Site 9 is approximately 100 m below the contact with the Valle de

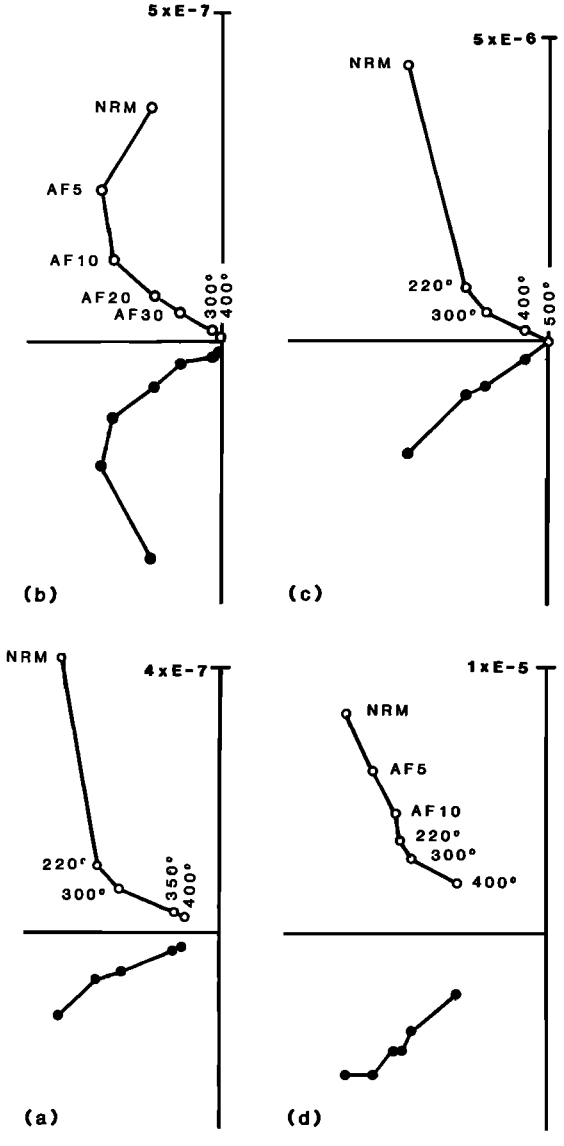


Fig. 6. Orthogonal vector projections for samples from the Atima Formation at site (a) A1, (b) A2, (c) A2, and (d) A3. Open circles are in the N-E-S-W plane and solid circles in the up-E-down-W plane. The axes are in (emu/gm = 2.7 x 10^3 A/m) and the demagnetization steps in milliteslas and degrees Celsius.

Angeles red beds and site 10 lies within 5 m of the contact. Examples of the sample response to demagnetization are shown in Figure 8.

Valle de Angeles Group: Lower Red Beds (V1-V10). The basal 500 m of the Valle de Angeles red beds were collected along Rio Guajiniquil starting at about 100 m above the contact with the Atima Formation (sites 1-3). All samples were extremely stable to thermal demagnetization and contained only one component of magnetization. A fourth site yielded a distinctively different direction (Figure 9) and is probably a slump block. It has been omitted from further consideration.

Our earlier work indicated that the lower Valle de Angeles red beds can be readily distinguished paleomagnetically from the upper red beds [Gose and Swartz, 1977]. With this in mind we collected undifferentiated red beds at two sites where the lithology suggested that they

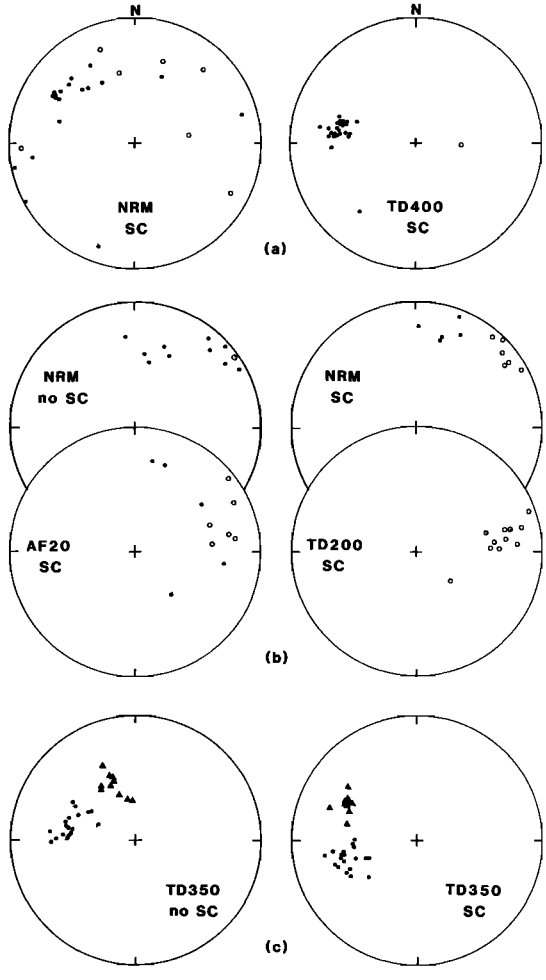


Fig. 7. Direction of magnetization of the Atima Formation: (a) site A4 at Buena Fe, (b) site A5 at Atima, and (c) sites A6 and A7 near Cedros. The triangles in Figure 7c represent a third site inferred to be a slump block.

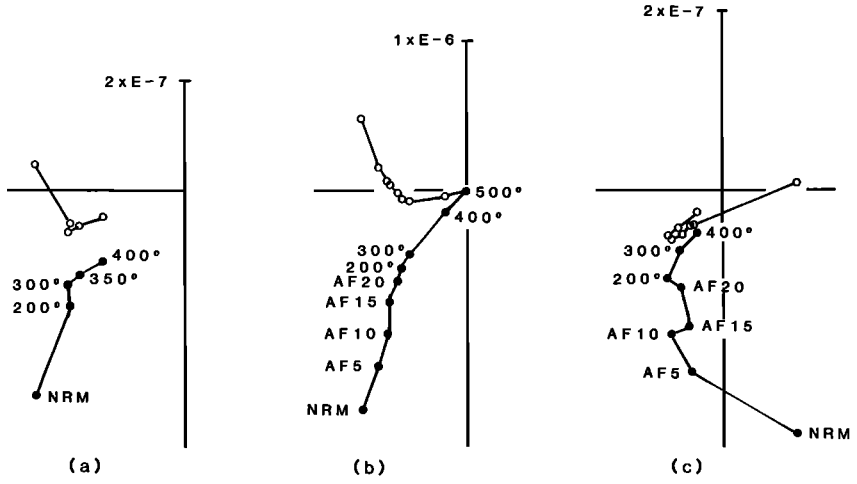


Fig. 8. Orthogonal vector projections of samples from the Atima Formation at sites A8, A9, and A10.

were within the lower sequence. Site 4 is in westernmost Honduras, 11.5 km north of Santa Rosa de Copan. Ten of 22 samples behaved erratically during demagnetization. These samples were from two poorly consolidated intervals where weathering has apparently destroyed the primary magnetization. The remaining samples yielded well-defined directions. The second site (site 5) is located near Salama, 270 km to the east. The samples from this site readily revealed the characteristic magnetization upon thermal demagnetization. Red beds which demonstrably are part of the lower Valle de Angeles Group, crop out south of Lake Yojoa [Curran, 1981]. Twelve samples from this locality (site 6) establish a tightly grouped mean direction. Examples of vector diagrams from sites 4-6 are shown in Figure 10.

The San Pedro Zacapa quadrangle, just to the southwest of Lake Yojoa, was mapped by Finch [1972, 1979]. Here, along Rio Salado, a long section of Valle de Angeles red beds is conformably overlain by the Cenomanian Jaitique Formation. Samples were collected at four sites (sites 7-10) spaced from the contact to several hundred meters down-section. Probably because of erosion in the river bed, none of the samples had a present field overprint. All four sites gave very similar results (Figure 11).

Valle de Angeles Group: Jaitique Formation (J1-J4). This formation consists of two members, a lower, thick-bedded limestone (100 m) and a thin-bedded shaly limestone (10 m) named the Guare member [Finch, 1981]. The basal 10 meters of the formation (site 1) were sampled along Rio Salado, site 2 near La Union spans the contact between the lower member and the Guare member, and site 3 (near La Union) and site 4 (near La Mission along highway 1) are both within the Guare member. All samples were first demagnetized by alternating fields to 15 or 20 mT, followed by thermal demagnetization. AF demagnetization reduced the intensities to 20-30% of

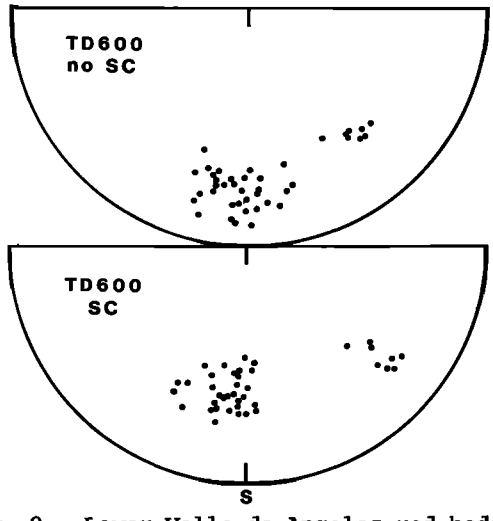


Fig. 9. Lower Valle de Angeles red beds along Rio Guajiniquil. Sites V1, V2, and V3 yield consistent directions. The fourth site, whose directions are displaced to the southeast, is interpreted to be slump block.

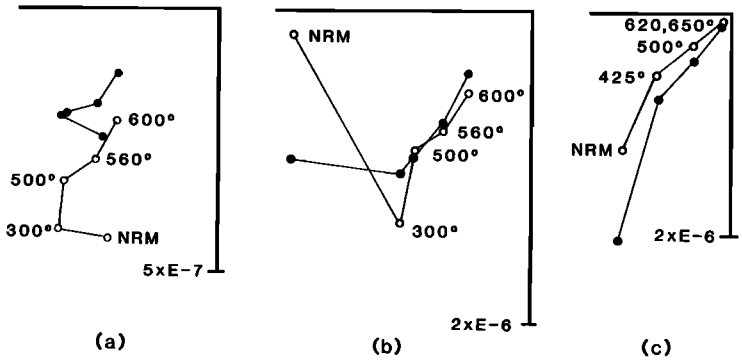


Fig. 10. Examples of As-Zijderveld diagrams from (a) site V4 near Santa Rosa de Copan, (b) site V5 near Salama, and (c) site V6 south of Lake Yojoa.

their NRM values but yielded only a moderate improvement in the directional scatter (Figure 12). By contrast, thermal demagnetization successfully removed the secondary magnetization, and the directions clustered well. Figure 12 also demonstrates the general observation that massive limestones (site 1), because of their resistance to weathering, usually do not acquire a present field magnetization.

Valle de Angeles Group: Esquias Formation (E1-E2). The Esquias Formation was sampled near its base (site 1) and top (site 2) at the type section [Horne et al., 1974]. All samples responded uniformly to demagnetization as shown in Figure 13. Changing from AF to thermal demagnetization caused a directional change of the mean by 3° for site 1 and 9° for site 2.

Valle de Angeles Group: Upper Red Beds (V11-V13). The upper red beds are well exposed along a new road from Tegucigalpa to Valle de Angeles. Two sites, about 200 m apart but stratigraphically equivalent, were sampled. The NRM directions exhibited a streak between the present field direction and a reversed magnetization. Upon heating, the directions of the samples from site 11 form a well-defined cluster (Figure 14a), whereas some samples from the second site (site 12) retained a present field overprint up to the highest demagnetization step of 665°C (Figure 14b).

Some 25 km east of these sites, we sampled an outcrop (site 13) where the red beds are overlain with angular unconformity by Tertiary volcanics. These samples contain a univectorial magnetization, antipodal to the two previous sites (Figure 14c).

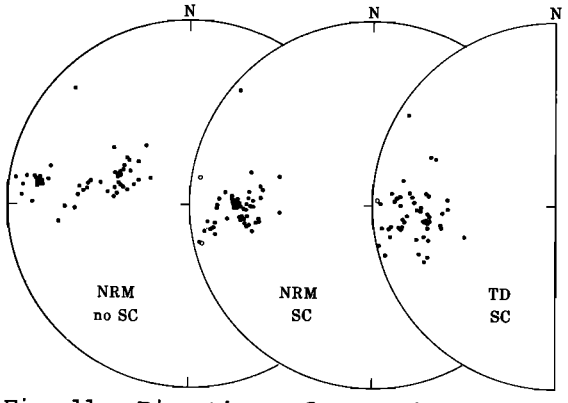


Fig. 11. Directions of magnetization of four sites (V7-V10) in the Lower Valle de Angeles red beds along Rio Salado. Site V7 is shown after thermal demagnetization to 500°C, the other three after 620°C. Note the positive fold test.

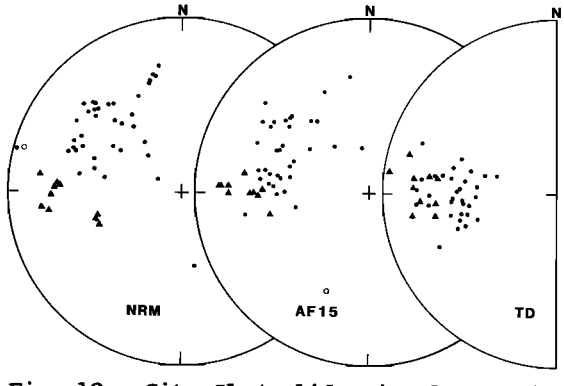


Fig. 12. Site J1 (solid triangles) and J2-J4 (solid circles) of the Jaitique Formation. Thermally demagnetized to 300°C (site J1), 400°C (sites J2 and J3), and 200°C (site J4).

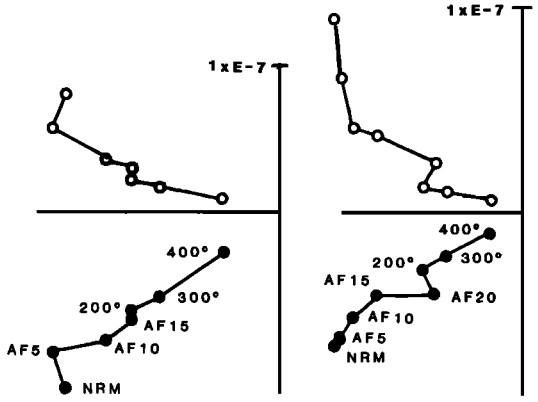


Fig. 13. Exemplary As-Zijderveld diagrams from the Esquias Formation (site E1 and E2).

Dated Intrusive Rocks

We collected paleomagnetic samples from four dated intrusive rocks in the hope that these results could be used to place constraints on the age ranges of some of the sedimentary units. Because of the lack of field evidence to the contrary, it is assumed that these intrusives were not deformed or tilted after emplacement.

The Dipilto batholith is a large granite along the Honduran-Nicaraguan border. A whole-rock Rb-Sr age of 140+ 15 m.y. (G. Horne, unpublished data, 1984) was obtained from four samples collected along the new highway near Dipilto about 11 km south of the border. At the same location (site DP1) we drilled 12 samples, only 5 of which contained a stable direction (Figure 15a).

The San Ignacio adamellite is a small stock in central Honduras. Horne et al. [1976] reported a K-Ar age of 114+1.8 m.y. from a biotite concentrate which they interpreted to be a minimum age. F. McDowell (unpublished data, 1984) obtained an age of 117.7 ±1.8 m.y. from biotites and 122.7 ±1.8 m.y. from hornblende. All but three samples of eleven paleomagnetic samples were directionally stable (site SI1, Figure 15b).

The nearby Minas de Oro granodiorite yielded a K-Ar age of 60.6 ± 1.3 m.y. from a biotite separate [Horne et al., 1976]. F. McDowell (unpublished data, 1984) obtained an age of 59.3 ±0.9 m.y. for biotite and 55.0 ±1.1 m.y. for hornblende for the same intrusive. Samples from two sites (MO1, MO2) responded uniformly to demagnetization. Scattered NRM direction started to cluster after demagnetization to 5 mT. Changing to thermal demagnetization improved the grouping and resulted in a 10 to 80 fold intensity increase (Figure 15c). The magnetization vector that was erased between 5 mT and 200°C has a mean declination of 313° and inclination of 27° (α_{95} = 8.7°), which is exactly antiparallel to the high-temperature component (D = 137°, I = -30°, α_{95} = 4.8°). Apparently this intrusive started to cool during a reversed interval and reached the ambient temperature during the subsequent normal interval.

The San Francisco stock, a porphyritic dacite, crops out a few kilometers south of the Minas de Oro stock. Horne et al. [1976] obtained a K-Ar age of 58.7 ± 0.7 m.y. from a biotite concentrate. AF demagnetization to 30 mT reduced the intensities of magnetization of samples from two sites (SF1, SF2) to less than 20%, most to less than 10%, of their NRM value but did not isolate a stable direction. Subsequent thermal demagnetization up to 600°C reduced the intensities to typically 1% of their NRM value, but a stable magnetization could be identified in most samples (Figures 15d, 15e). It is doubtful, however, that the primary magnetization has been isolated. The San Francisco stock is very similar in age to the Minas de Oro stock and the two bodies should have the same direction of magnetization (+180°), which they do not (Table 1 and Figure 16). Rather, the two pole positions from the San Francisco stock are displaced from the Minas de Oro poles

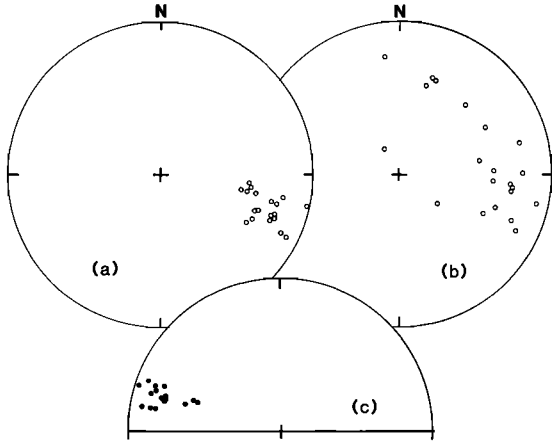


Fig. 14. Directions observed in the upper Valle de Angeles red beds (sites V11-V13). Note the reversal.

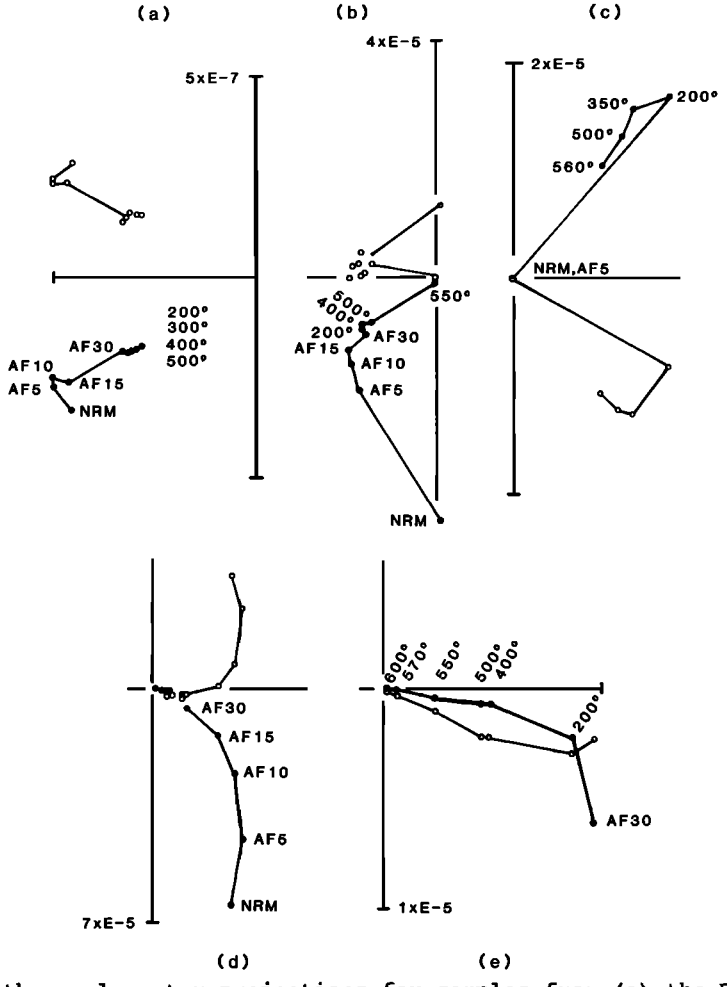


Fig. 15. Orthogonal vector projections for samples from (a) the Dipilto, (b) San Ignacio, (c) Minas de Oro, (d) and (e) San Francisco intrusive. Figure 15e is an enlargement of Figure 15d.

along the great circle which also contains the Minas de Oro pole and antipole. This suggests that the San Francisco samples contain two components of magnetization, one parallel and one antiparallel to the Minas de Oro direction. The fact that only one component was resolved during demagnetization is explained by the observation that the San Francisco dacite contains both primary and secondary magnetite [Horne et al., 1976] which, seemingly, have very similar demagnetization characteristics. Clearly, both components must be of very similar age which implies that the alteration (chloritization) of the San Francisco stock must have occurred soon after intrusion and that the radiometric age determination closely indicates the time of crystallization. These arguments also support the validity of the Minas de Oro results.

VALIDITY OF DATA

The paleomagnetic results are summarized in Table 1. All listed sites yielded statistically good data as evidenced by their small circles of confidence (α_{95}) and large precision parameters (k). It needs to be demonstrated, though, that the primary magnetizations have been isolated and that the results can be used for regional tectonic interpretations. The pole positions are far removed from the geographic north pole regardless of whether the data are structurally corrected. Thus the magnetization is not of recent origin. The large spread in pole postions rules

TABLE 1. Summary of Paleomagnetic Data From the Chortis Block

	TREAT	N	R	D	I	k	α_{95}	COLAT	LAT	LONG	dp	dm
						Sedimentary Rocks						
Todos Santos Fm												
TS1	TD500	44/5	42.9	294.1	31.0	38.9	3.5	73.3	26.9	193.6	2.2	3.9
TS2	TD500	27/7	26.5	292.5	25.5	54.4	3.8	76.6	24.8	190.2	2.2	4.1
Cantarranas Fm												
C1	ORTHO	20/18	19.4	283.2	33.0	30.2	6.0	72.0	16.7	196.7	3.9	6.8
Atima Fm												
A1	ORTHO	14/3	13.8	281.5	26.3	54.1	5.5	76.1	14.4	192.8	3.2	5.9
A2	ORTHO	37/19	35.8	296.2	39.2	30.5	4.3	67.8	29.4	199.4	3.1	5.2
A3	TD400	20/0	19.9	297.4	33.1	245.4	2.1	72.0	30.1	194.6	1.3	2.4
A4	TD400	24/2	23.8	280.8	38.4	138.2	2.5	68.4	15.2	200.8	1.8	3.0
A5	ORTHO	11/11	10.8	82.5	-30.1	63.9	5.8	73.8	-2.8	199.1	3.5	6.4
A6	TD350	8/1	8.0	255.9	34.5	147.5	4.6	71.0	-8.1	205.0	3.0	5.3
A7	TD350	10/0	9.9	252.8	46.2	71.1	5.8	62.5	-7.9	214.1	4.7	7.4
A8	TD350	19/6	18.5	243.4	46.6	38.9	5.4	62.1	-15.4	217.4	4.5	7.0
A9	TD300	14/3	13.9	226.8	39.0	112.0	3.8	67.9	-31.3	219.8	2.7	4.5
A10	ORTHO	12/1	11.5	226.7	50.0	20.1	9.9	59.2	-26.1	227.8	8.9	13.3
Valle de Angeles Group												
Lower Red Beds												
V1	TD600	15/0	14.8	190.6	39.0	82.8	4.2	67.9	-51.8	255.9	3.0	5.0
V2	TD650	10/0	9.8	191.1	37.1	47.3	7.1	69.3	-53.0	254.5	4.9	8.3
V3	TD600	8/0	8.0	181.9	33.3	297.8	3.2	71.8	-57.1	268.6	2.1	3.7
V4	TD600	12/10	11.5	219.7	21.7	22.8	9.3	78.7	-42.8	212.6	5.2	9.8
V5	TD600	15/4	14.8	221.4	32.0	80.0	4.3	72.7	-38.0	220.2	2.7	4.8
V6	ORTHO	12/6	11.9	219.0	35.8	84.4	4.8	70.1	-38.4	223.0	3.2	5.5
V7	TD620	14/0	13.9	269.1	19.5	96.5	4.1	79.9	1.7	191.8	2.2	4.2
V8	ORTHO	11/2	10.8	265.1	24.5	59.5	6.0	77.2	-1.4	195.5	3.4	6.4
V9	TD600	15/0	14.8	260.9	30.2	91.3	4.0	73.8	-4.3	199.9	2.5	4.5
V10	TD500	10/4	9.7	262.3	12.7	35.7	8.2	83.6	-5.8	190.0	4.3	8.3
Jaitique Fm												
J1	TD300	10/2	9.8	272.4	22.1	43.6	7.4	78.5	5.1	192.3	4.1	7.8
J2	TD400	10/1	9.9	264.8	47.5	76.2	5.6	61.4	2.7	210.4	4.7	7.2
J3	TD400	11/0	10.9	275.1	42.7	89.3	4.9	65.3	10.7	204.5	3.7	6.0
J4	TD200	14/2	13.5	267.5	33.4	24.8	8.1	71.8	2.3	200.3	5.3	9.3
Esquias Fm												
E1	TD300	18/3	17.9	283.8	34.5	224.4	2.3	71.0	17.5	198.3	1.5	2.7
E2	TD300	17/1	16.9	280.4	34.1	208.1	2.5	71.3	14.3	198.5	1.6	2.8

TABLE 1. (continued)

	TREAT	N	R	D	I	k	α_{95}	COLAT	LAT	LONG	dp	dm
Upper Red Beds												
V11	TD620	18/0	17.5	107.1	-28.5	36.6	5.8	74.8	19.8	194.2	3.5	6.4
V12	ORTHO	17/5	16.4	104.2	-25.3	25.8	7.2	76.7	16.7	192.8	4.1	7.7
V13	TD620	16/0	15.7	285.7	19.5	44.6	5.6	80.0	17.5	189.5	3.0	5.8
Igneous Rocks												
Dipilto Granodiorite (140 my)												
DP1	ORTHO	5/7	4.9	299.3	25.5	51.9	10.7	76.6	31.2	191.1	6.2	11.6
San Ignacio Adamellite (118 my)												
SI1	ORTHO	8/3	7.9	266.6	33.7	70.7	6.6	71.6	1.5	201.2	4.3	7.6
Minas de Oro Granodiorite (59 my)												
MO1	ORTHO	9/2	8.9	137.9	-31.5	86.8	5.6	72.9	49.5	191.7	3.5	6.2
MO2	ORTHO	9/1	8.8	136.6	-29.2	35.5	8.8	74.4	48.2	189.6	5.3	9.7
San Francisco Dacite (59 my)												
SF1	ORTHO	11/8	10.7	117.7	-10.0	35.6	7.8	85.4	29.8	179.4	4.0	7.8
SF2	ORTHO	11/2	10.9	117.1	11.3	91.9	4.8	84.3	24.4	169.2	2.5	4.9

TREAT, treatment (TD, thermal demagnetization at specified temperature; ORTHO, orthogonal vector projection); N, number of samples included/rejected in the analysis; R, resultant vector; D, declination; I, inclination; k, precision parameter; α_{95}, 95% circle of confidence; COLAT, paleo-colatitude; LAT, LONG, latitude and east-longitude of paleomagnetic pole; dp, dm, semi-angles of 95% ellipse of confidence.

out the possibility of remagnetization by a regional metamorphic event, and indeed no such an event has been reported in the geological literature. It is also clear that the magnetization predates the "Laramide" deformation. The four Valle de Angeles sampling sites along Rio Salado (sites V7-V10) yield a consistent direction of magnetization only after structural correction (Figure 11, Table 2), implying a prefolding age. No other fold test was possible on a local scale, but it can be applied on a regional scale. Table 2 compares the precision parameter k before and after applying tilt corrections for eight subsets of pole positions grouped according to their known or inferred age. In all cases, the data cluster dramatically better after structural correction, attesting to a predeformational magnetization. Note that several of the subsets in Table 2 have poles from different formations, i.e., lithologies.

Thermal demagnetization suggested that the stable remanence in the limestones is carried by magnetite. In order to verify this, six samples from the Atima Formation were given an isothermal remanent magnetization (IRM) in fields up to 0.5 T. Three of these samples had been previously thermally demagnetized to 400°C, whereas the other three were the untreated sister samples. There was no difference between the heated and unheated samples. They saturated between 0.15 and 0.20 T, indicating the presence of magnetite, further evidence that the limestones retained their primary magnetization.

It also needs to be shown that the pole positions are representative for some larger area and are not the result of local tectonics. This is particularly

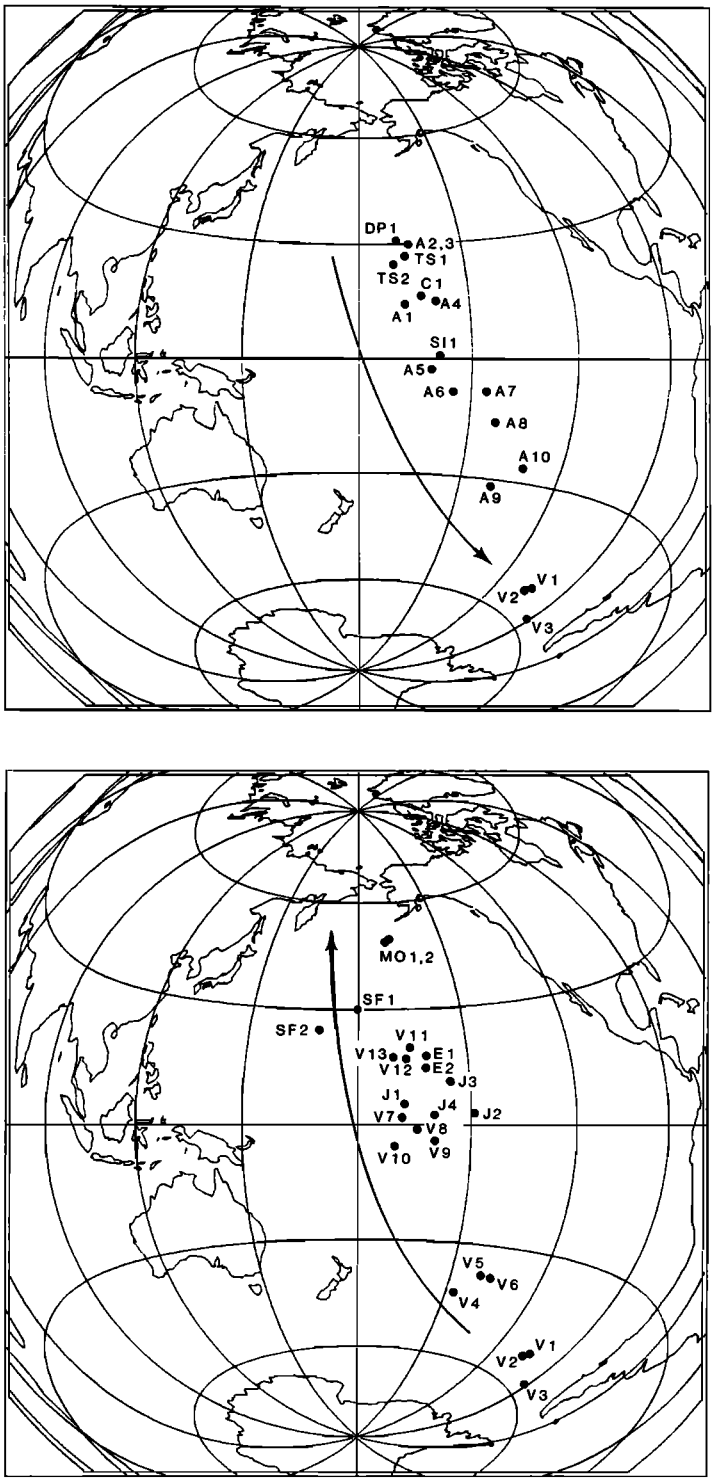


Fig. 16. Pole positions from the Chortis Block. Large arrows indicate age trend from old to young. Sites are coded to Table 1.

TABLE 2. Comparison of Fisher Precision Parameter K for Groups of Pole Positions Before (K_b) and After (K_a) Structural Corrections

Sites	N	K_b	K_a
T1-A4	7	61.5	111.2
A5-A7, SI1	4	25.8	101.7
A8-A10	3	13.7	72.7
V1-V3	3	183.3	237.5
V4-V6	3	21.6	275.6
V7-V10	4	31.6	218.3
J1-J4	4	19.3	90.8
E1-V13	5	33.0	381.1

N is the number of poles included in each group.

important for the data presented here because the pole positions lie along a small circle of radius 72.3° $+5.2^{\circ}$ centered at 18.0°N, 272.2°E, i.e., the center falls within a few degrees of the sampling area (Figure 16). Such a distribution can be the result of supracrustal rotation of small blocks and/or incorrect tilt corrections, as MacDonald [1978, 1980] pointed out. These apparent rotations increase with the dip of the strata but cannot be larger than the dip angle except in the case of multiple deformations. The dips of the sedimentary rocks in this study vary from 12° to 45° with a mean of 26°. Thus any error due to incorrect tilt corrections is accordingly small and will not change the distribution of pole positions in a major way.

The validity of the data for regional interpretation is further supported by considering the sampling site distribution relative to the major structural features in Honduras (Figure 3). Most of these faults and associated grabens are of Late Tertiary age and are still active [Muehlberger, 1976; Mann and Burke, 1984]. The paleomagnetic data are internally consistent irrespective of their site locations (Tables 1 and 2, Figure 16). For example, the results from the Atima Formation at site 5 in western Honduras closely agree with the data from sites 6 and 7 on the east side of the Honduran Depression, and all three sites yielded pole positions very similar to the pole of the approximately coeval San Ignacio intrusive. Sites 4, 5, and 6 of the lower Valle de Angeles red beds, separated by some 270 km, span all major structures, and the Jaitique Formation was sampled within, to the west, and to the east of the Santa Barbara Graben. Both data sets are internally consistent. For such a consistency to be the result of local rotations would indeed be most fortuitous.

POLARITY OF MAGNETIZATION

Before the data can be used for tectonic interpretations, it is mandatory to establish the polarity of magnetization, for it determines the sense of rotation and the paleolatitude. Most of the sampled rock units are of Aptian through Cenomanian age, i.e., they were deposited during a period of constant normal magnetic polarity (84-118 m.y., Harland et al. [1982]). All but three of the sedimentary sites are of the same polarity, which thus must be normal.

One reversed horizon was encountered in the Atima Formation at its type locality (site A5). Its paleontological age of Upper Aptian (113-115 m.y.) suggests that this reversal may correspond to seafloor magnetic anomaly M0 (118-119 m.y., Harland et al. [1982]). The other two reversed sites are from one stratigraphic horizon in the upper Valle de Angeles red beds (sites V11, V12) which could correspond to any of the reversals younger than the long normal interval.

The choice of polarity implies that the Paleocene Minas de Oro and San Francisco stocks are reversely magnetized. Indeed, only this polarity assignment will place the Chortis Block in a position that is compatible with other data as discussed in the next section. The agreement between the observed and expected polarities is further proof that the rocks retained their primary magnetization.

DISCUSSION

The pole positions depicted in Figure 16 cover the age range from 140 to 60 m.y., that is, the entire Cretaceous and Early Tertiary. They vary in a dramatic but smooth fashion and thus are well suited for stratigraphic correlations which is the subject of another paper [W. A. Gose and R. C. Finch, manuscript, in preparation, 1985]. I only want to point out that the sequence of pole positions is

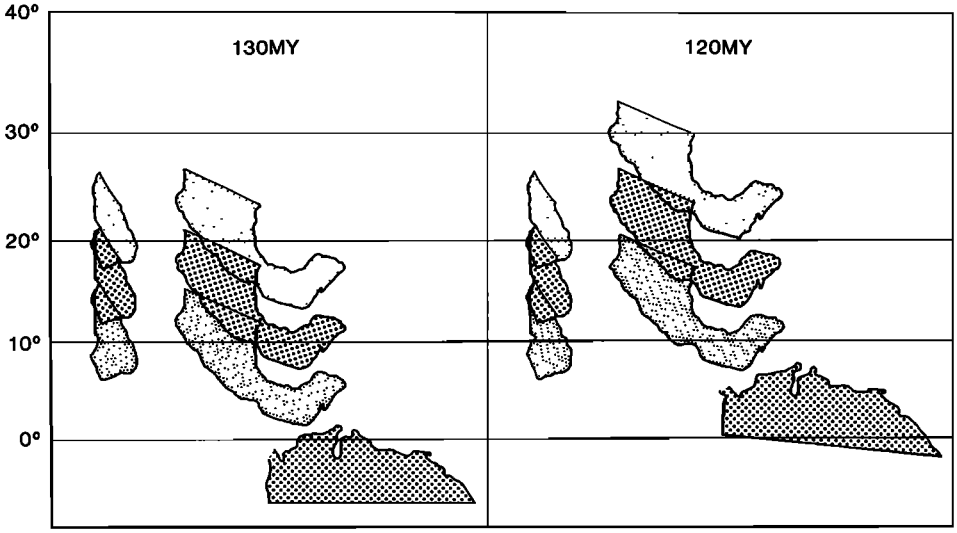


Fig. 17. Illustration of latitudinal error for the position of the Chortis Block due to (1) error of the Honduran data, (2) error of the North American reference data, and (3) uncertainty in age of the Honduran sites.

in full agreement with the known stratigraphy of Honduras [Finch, 1981]. The polar wander path presented here is very similar to the one published by Gose and Swartz [1977] but is based on more than 3 times the number of sites. The only major change results from a revised pole position for the Cantarranas Formation and basal Atima Formation as discussed earlier.

Before using the pole positions for plate reconstructions, it is important and rather instructive to consider the accuracies of the Honduran data, the reference data for North America, and the age assignments. The largest uncertainty is associated with the Lower Cretaceous (poles TS1 to A4). Geologically most significant is the latitudinal position of the Chortis Block relative to North and South America. The latitude of the Chortis Block has an error of $\pm 5.7^{\circ}$ (95% confidence limit). Because the age can only be specified as Lower Cretaceous (but pre-Albian), one may use either the 130 m.y. or the 120 m.y. reference poles for North America [Harrison and Lindh, 1982]. Figure 17 depicts the positions of Chortis relative to North America (only southern Mexico is shown) for both times together with the position of South America [Ladd, 1976], assumed to be precisely known. Using the extremes of the error limits, Honduras may have been located as far north as northernmost Mexico or as far south as northern South America.

This example represents the worst case for the data under consideration. Even if the inaccuracy of the age assignment could be eliminated, a 5° error in the paleomagnetic data, which is a respectably small error, becomes significant when one realizes that the width of the Caribbean is only about 10°. In contrast to the latitudinal error, the rotational error is of little concern for these data because of the magnitude of the paleomagnetically implied rotations (see Figure 18). It is clear, however, that paleomagnetic data alone do not have the resolving power to delineate in detail the tectonic evolution of an area as small as the Caribbean. In the forthcoming discussion, I will first describe the motion of the Chortis Block as indicated by the paleomagnetic data and then present other data which bear on this problem.

Figure 18 depicts the position of the Chortis Block relative to North America. The position of North America has been calculated for discrete times by using the polar wander curve of Harrison and Lindh [1982]. These times were chosen to most closely approximate the ages of the Honduran rocks. The position of the Chortis Block at 140 m.y. was derived from pole DP1, the positions from 130 to 90 m.y. from the means of site means as grouped in Table 2, and the 60 m.y. position from poles MO1 and MO2. For visual reference, Chortis is shown close to North America in all diagrams, but it should be kept in

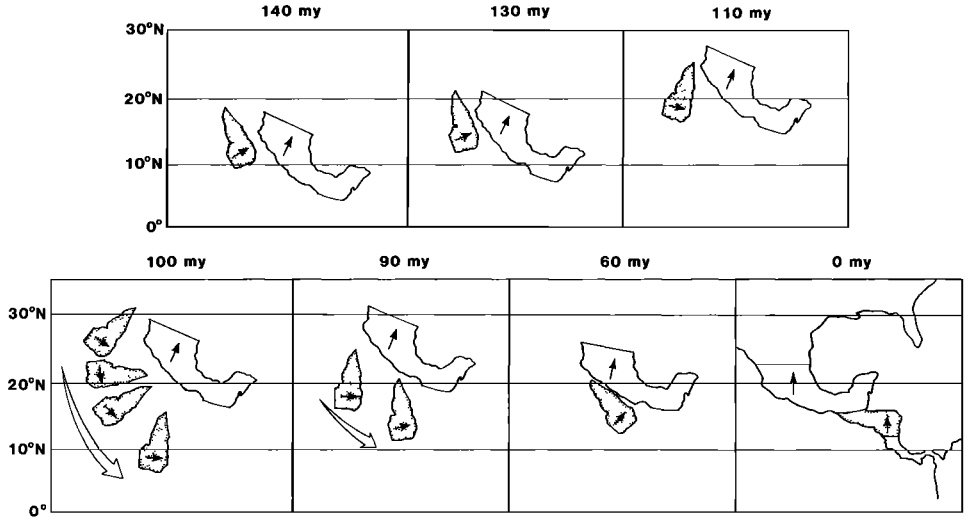


Fig. 18. Position of the Chortis Block relative to North America. The position of North America has been calculated from the polar wander curve of Harrison and Lindh [1982] for discrete times. These times were chosen to most closely correspond to the ages of the Honduran rocks. The longitudinal relationship is not constrained by the paleomagnetic data.

mind that the longitudinal relationship is unconstrained by paleomagnetic data.

The oldest data point (140 m.y.) places the Chortis Block at 13° N latitude, using the present coordinates of Tegucigalpa (14°N, 273°E) as the point of reference. Chortis was in a position rotated 61° clockwise with respect to geographic north and 37° clockwise with respect to North America. During the Lower Cretaceous, the Chortis Block moved seven degrees north and rotated to 100° clockwise (130 and 110 m.y.). In Late Albian (?) time (lowermost Valle de Angeles), the Chortis Block was rotated 171° (149° relative to North America). At about 100 m.y., the rotation changed from clockwise to counterclockwise. At 60 m.y., the Chortis Block was at latitude 16°N, rotated 32° clockwise relative to North America. It is worth pointing out that while the rotations are rather large, the rate of rotation averages only about 3° per million years for both the clockwise and counterclockwise motion. Very similar values have been obtained for the displaced terranes of the Oregon-Washington coast, ranging from 1.5°/m.y. [Beck and Plumley, 1980] to 6°/m.y. [Magill et al., 1981]. The latitudinal displacements are minor and may not be significant when considering the error limits of the paleomagnetic data and the uncertainties of the age assignments (see earlier discussion and Figure 17).

The Early Cretaceous position of the Chortis Block is consistent with the suggestion that it may have been attached to Mexico. Such a configuration has been proposed by several authors [e.g., Anderson and Schmidt, 1983; Aubouin et al., 1982; Pindell and Dewey, 1982; Salvador and Green, 1980], but it should be emphasized, as most of these authors acknowledge, that the pre-Cretaceous geology of Honduras as well as of southern Mexico is too poorly known to permit an evaluation [Finch, 1981; Ortega-Gutierrez, 1981].

After undergoing large rotations, the Chortis Block is in a position by the end of the Cretaceous which suggests that it had been sutured onto southern Mexico/Guatemala. This interpretation is strongly supported by geochemical data. Lawrence [1975] demonstrated that the majority of rocks of the El Tambor Formation which crops out in the Motagua Valley, the present plate boundary, are normative tholeiites of mid-ocean ridge affinity which underwent regional metamorphism, probably in the Late Cretaceous. Sutter [1979] reports several argon-argon ages from the Chuacus Series which interleaves with the El Tambor Formation. He interprets the metamorphic age of about 70 m.y. as "the closest approximation to the time of collision at this point along the Motagua zone." (Note: As depicted in Figure 18, the Chortis Block is not in contact with the area in the Motagua Valley dis-

cussed by Lawrence [1975] and Sutter [1979]. However, if one includes part of the Nicaraguan Rise, as discussed later, the two areas are in juxtaposition.)

For the Cenozoic, these paleomagnetic data predict a left-lateral strike-slip motion and an associated 32° counterclockwise rotation with respect to North America. Preliminary paleomagnetic data by MacDonald [1976] seem to support this inference. In a recent paper on Cenozoic rifts in the northern Caribbean, Mann and Burke [1984] presented a Paleocene reconstruction that places the Chortis Block at the very same position as advocated in this paper (compare their Figure 4).

In the foregoing discussion it was assumed that the paleomagnetic data pertain to the motion of only the Chortis Block. Today, this block is part of the Caribbean plate, and it therefore needs to be evaluated whether or not it was part of the Caribbean plate during the Cretaceous; or, put differently, do the paleomagnetic data from Honduras reflect the motion of the entire Caribbean plate? The area north of the Motagua fault, the present plate boundary, clearly did not partake in this motion as evidenced by the paleomagnetic results from northern Guatemala [W. A. Gose and R. C. Finch, manuscript, in preparation, 1985] and nearby Mexico [Guerrero, 1975]. Results from Upper Cretaceous and Lower Tertiary rocks from southernmost Nicaragua and Costa Rica [Gose, 1983] are distinctly different from the Honduran data and suggest that the boundary between the Chortis Block and southern Central America probably coincides with the eastern edge of the Nicaraguan Depression.

Jamaica could possibly be structurally connected to the Chortis Block via the Nicaraguan Rise. However, new paleomagnetic results from Jamaica [Gose and Testarmata, 1983] show no evidence for rotation of this island during the Cretaceous, and Jamaica was, therefore, not part of the Chortis Block. The eastern limit of the Chortis Block is probably delineated by a line striking N15°E through the San Andres and Providencia islands, characterized by a series of normal faults [Case and Holcombe, 1980]. The same line can also be recognized on seismic epicenter maps [e.g., Sykes et al., 1982, Gose, 1985]. In a detailed study of marine single and multichannel seismic data Bowland [1984] demonstrated recent tectonic activity along this boundary and argued that it separates two different basements.

It thus seems clear that the Chortis Block is a distinct tectonic entity within the present Caribbean plate. Gose and Swartz [1977] and Gose [1981] pointed out that it would indeed be most difficult to reconcile the large rotations of Chortis with the existence of a rigid Caribbean plate, at least in the Cretaceous, and suggested that this plate may not behave rigidly even today.

Paleomagnetic data from around the Caribbean (see Gose [1985], for a review and references) attest to a great mobility within the plate margin. Large rotations, often about 90°, have been found along the northern as well as southern Caribbean plate boundaries implying broad plate boundary zones. The same conclusion has also been reached by Burke et al. [1980] and Sykes et al. [1982]. Although the paleomagnetic data base is still too incomplete for a detailed model, some general features are apparent. Most of the rotations occurred in post-Cretaceous, often post-Eocene time, and are clockwise in the southern plate boundary zone and counterclockwise in the northern zone. These rotations seem to be directly related to the eastward motion of the Caribbean plate since the Oligocene [e.g., Wadge and Burke, 1983]. Marine seismic, gravity, and magnetic data from the Caribbean suggest that this plate has also undergone substantial internal deformation during the Cenozoic [Diebold et al., 1981; Lu and McMillen, 1982; Wadge and Burke, 1983; Bowland, 1984; Mann and Burke, 1984].

The analogy that comes to mind is that of ice rafts on a river. The overall motion is downstream, but individual rafts are free to rotate with a preference of counterclockwise rotation along the left bank (left-lateral shear) and clockwise rotation along the right bank. If this picture is conceptually correct, then great caution needs to be exercised when using present geometry and plate motion parameters to infer the tectonic evolution of the Caribbean region.

EPILOGUE

The plate tectonic setting of the Pacific basin prior to ~40 m.y. and particularly prior to 85 m.y. is poorly constrained and recontructions critically

depend on the assumed location and orientation of spreading centers active at the time. Using the relative plate motion circuit approach to calculate the rate and direction of the subduction of the Farallon plate beneath North America, Jurdy [1984] shows that the direction of subduction between 60 and 40 m.y. could be either north easterly or easterly, depending on whether Antarctica is treated as a single plate or as two plates. According to Page and Engebretson [1984], the area off Mexico may have been part of either the Kula plate or Farallon plate at 74 m.a. If this area were part of the Kula plate, subduction would result in right-lateral shear; if part of the Farallon plate, the shear would be sinistral. For older times the inferred plate motions are even more model dependent. From the analysis of hot spot tracks, Duncan and Hargraves [1984] propose a northeasterly motion of the Farallon plate at 140 m.y. whereas a southeasterly motion is advocated by Pindell [1985] for the same time to account for possible left-lateral shear zones in Mexico.

It seems reasonable to assume that the large rotations of the Chortis Block during the Cretaceous are the result of oblique subduction and the ensuing shear couple. The Lower Cretaceous clockwise rotation of Chortis favors those models which yield a northerly to northeasterly convergence for that time period. At about 100 m.y., the sense of Chortis rotation changed to counterclockwise which seems to be related to the major plate reorganization occurring at that time. North and South America had separated enough to allow "Pacific" crust to enter the Caribbean region [e.g., Pindell and Dewey, 1982; Duncan and Hargraves, 1984]. This new plate arrangement would subject the Chortis Block to left-lateral shear resulting in the observed counterclockwise rotation.

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REFERENCES

Anderson, T. H., and V. A. Schmidt, The evolution of Middle America and the Gulf of Mexico-Caribbean Sea region during Mesozoic time, Geol. Soc. Am. Bull., 94, 941-966, 1983.

Aubouin, J., J. Azema, J.-Ch. Carfantan, A. Demant, C. Rangin, M. Tardy, and J. Tournon, The Middle America trench in the geological framework of Central America, in Aubouin, J., R. von Huene et al., Init. Repts. Deep Sea Drill. Proj., 67, 351-382, 1982.

Beck, M. E., and P. W. Plumley, Paleomagnetism of intrusive rocks in the Coast Range of Oregon: Microplate rotations in middle Tertiary time, Geology, 8, 575-577, 1980.

Bonis, S., O. H. Bohnenberger, and G. Dengo, Mapa geologico de la Republica de Guatemala, scale 1:500,000, Inst. Geogr. Nac., Guatemala City, 1970.

Bowland, C., Seismic stratigraphy and structure of the western Colombian basin, Caribbean Sea, Master's thesis, 247 pp., Univ. of Tex. at Austin, Austin, 1984.

Burke, K., J. Grippi, and A. M. C. Sengor, Neogene structures in Jamaica and the tectonic style of the northern Caribbean plate boundary zone, J. Geol., 88, 375-386, 1980.

Case, J. E., Oceanic crust forms basement of eastern Panama, Geol. Soc. Am. Bull., 85, 645-652, 1974.

Case, J. E., and T. L. Holcombe, Geologic-tectonic map of the Caribbean region, scale 1:2,500,00 , Misc. Invest. Ser., map I-1100, U.S. Geol. Surv., Reston, Va., 1980.

Curran, D. W., Geologic map of Honduras, Taulabe quadrangle, scale 1:50,000, Hoja 2660 III G, Inst. Geogr. Nac., Tegucigalpa, Honduras, 1981.

de Boer, J., Mapa geofisico preliminar de Costa Rica, scale 1:500,000, Inst. Geogr. Nac., San Jose, Costa Rica, 1974.

de Boer, J., The outer arc of the Costa Rica orogen (oceanic basement complexes of the Nicoya and Santa Elena peninsulas), Tectonophysics, 56, 221-259, 1979.

Dengo, G., Problems of tectonic relations between Central America and the Caribbean, Trans. Gulf Coast Assoc. Geol. Soc., 19, 311-320, 1969.

Dengo, G., and O. Bohnenberger, Structural development of northern Central America, Mem. Am. Assoc. Pet. Geol., 11, 203-220, 1969.

Diebold, J. B., P. L. Stoffa, P. Buhl, and M. Truchan, Venezuela Basin crustal structure, J. Geophys. Res., 86, 7901-7923, 1981.

Duncan, R. A., and R. B Hargraves, Plate tectonic evolution of the Caribbean region in the mantle reference frame, Mem. Geol. Soc. Am., 162, 81-93, 1984.

Elvir, A. R., Mapa geologico de la Republica de Honduras, scale 1:500,000, Inst. Geogr. Nac., Tegucigalpa, Honduras, 1974.

Finch, R. C., Geology of the San Pedro Zacapa quadrangle, Honduras, Central America, Ph.D. dissertation, 238 pp., Univ. of Tex. at Austin, Austin, 1972.

Finch, R., Geologic map of Honduras, San Pedro Zacapa quadrangle, scale 1:50,000, Hoja 2560II, Inst. Geogr. Nac., Honduras, 1979.

Finch, R., Mesozoic stratigraphy of central Honduras, Am. Assoc. Pet. Geol. Bull., 65, 1320-1333, 1981.

Gose, W. A., The Caribbean collage, paper presented at meeting, Int. Assoc. Geomagn. Aeron., Edinburgh, 1981.

Gose, W. A., Late Cretaceous-Early Tertiary tectonic history of southern Central America, J. Geophys. Res., 88, 10,585-10,592, 1983.

Gose, W. A., Caribbean tectonics from a paleomagnetic perspective, The Great American Biotic Interchange, Topics in Geobiology, edited by F. G. Stehli and S. D. Webb, Plenum Press, in press, 1985.

Gose, W. A., and M. M. Testarmata, Paleomagnetic results from sedimentary rocks in Jamaica--Initial results, J. Geol. Soc. Jamaica, 22, 16-24, 1983.

Gose, W. A., and D. K. Swartz, Paleomagnetic results from Cretaceous sediments in Honduras: Tectonic implications, Geology, 5, 505-508, 1977.

Gose, W. A., R. C. Finch, and G. S. Horne, Paleomagnetic results from Cretaceous sediments in Honduras: Tectonic implications: Reply, Geology, 6, 444-447, 1978.

Guerrero, J. C., Contributions to paleomagnetism and Rb-Sr geochronology, Ph.D. thesis, 128 pp., Univ. of Tex. at Dallas, Dallas, 1975.

Harland, W. B., A. V. Cox, P. G. Llewellyn, C. A. G. Pickton, A. G. Smith, and R. Walters, A Geologic Time Scale, 131 pp., Cambridge University Press, New York, 1982.

Harrison, C. G. A., and T. Lindh, A polar wandering curve for North America during the Mesozoic and Cenozoic, J. Geophys. Res., 87, 1903-1920, 1982.

Horne, G. S., M. G. Atwood, and A. P. King, Stratigraphy, sedimentology, and paleoenvironment of Esquias Formation of Honduras, Am. Assoc. Pet. Geol. Bull., 58, 176-188, 1974.

Horne, G. S., P. Pushkar, and M. Shafiqullah, Preliminary K-Ar age data from the Laramide Sierras of central Honduras, Informe IV Reunion de Geologos de America Central, Publ. Geol. ICAITI, 5, 92-98, 1976.

Jurdy, D. M., The subduction of the Farallon plate beneath North America as derived from relative plate motions, Tectonics, 3, 107-113, 1984.

King, A. P., Geologic map of Honduras, Cedros quadrangle, scale 1:50,000, Inst. Geogr. Nac., Tegucigalpa, Honduras, 1973.

Ladd, J. W., Relative motion of South America with respect to North America and Caribbean tectonics, Geol. Soc. Am. Bull., 87, 969-976, 1976.

Lawrence, D. P., Petrology and structural geology of the Sanarate-El Progreso area, Guatemala, Ph.D. thesis, 255 pp., State Univ. of N.Y., Binghamton, 1975.

Lu, R. S., and K. J. McMillen, Multichannel seismic survey of the Colombia basin and adjacent margins, Studies in Continental Margin Geology, edited by J. S. Watkins, and C. L. Drake, Mem. Am. Assoc. Pet. Geol., 34, 395-410, 1982.

MacDonald, W. D., Paleomagnetic results from Honduras, Central America, and the derivation of the Caribbean plate (abstract), Eos Trans. AGU, 57, 239, 1976.

MacDonald, W. D., Paleomagnetic results from Honduras: Tectonic implications: Comment, Geology, 6, 443-444, 1978.

MacDonald, W. D., Net tectonic rotation, apparent tectonic rotation, and the

structural tilt correction in paleomagnetic studies, J. Geophys. Res., 85, 3659-3669, 1980.
Magill, J. R., A. V. Cox, and R. Duncan, Tillamook volcanic series: Further evidence for tectonic rotation of the Oregon Coast Range, J. Geophys. Res., 86, 2953-2970, 1981.
Mann, P., and K. Burke, Cenozoic rift formation in the northern Caribbean, Geology, 12, 732-736, 1984.
Mills, R. A., K. E. Hugh, D. E. Feray, and H. C. Swolfs, Mesozoic stratigraphy of Honduras, Am. Assoc. Pet. Geol. Bull., 51, 1711-1786, 1967.
Molnar, P., and L. R. Sykes, Tectonics of the Caribbean and Middle America region from focal mechanisms and seismicity, Geol. Soc. Am. Bull., 80, 1639-1684, 1969.
Muehlberger, W. R., The Honduras depression, Informe IV Reunion de Geologos de America Central, Publ. Geol. ICAITI, 5, 43-51, 1976.
Ortega-Gutierrez, F., Metamorphic belts of southern Mexico and their tectonic significance, Geofis. Int., 20, 177-202, 1981.
Page, B. M., and D. C. Engebretson, Correlation between the geologic record and computed plate motions for Central California, Tectonics, 3, 133-155, 1984.
Pindell, J. L., Alleghenian reconstruction and subsequent evolution of the Gulf of Mexico, Bahamas, and proto-Caribbean, Tectonics, 4, 1-39, 1985.
Pindell, J., and J. F. Dewey, Permo-Triassic reconstruction of western Pangea and the evolution of the Gulf of Mexico/Caribbean region, Tectonics, 1, 179-211, 1982.
Salvador, A., and A. R. Green, Opening of the Caribbean tethys, 26th International Geological Congress, Paris, Mem. BGRM, 115, 1980.
Schmidt-Effing, R., Alter und Genese des Nicoya Komplexes, einer ozeanischen Palaokruste (Oberjura bis Eozan) im sudlichen Zentral Amerika, Geol. Rundsch., 68, 457-494, 1979.
Schuchert, C., Historical Geology of the Antillean-Caribbean Region, 811 pp., John Wiley, New York, 1935.
Sutter, J. F., Late Cretaceous collisional tectonics along the Motagua fault zone, Guatemala (abstract), Geol. Soc. Am. Abs. with Programs, 11, 525, 1979.
Sykes, L. R., W. R. McCann, and A. L. Kafka, Motion of Caribbean plate during last 7 million years and implications for earlier Cenozoic movements, J. Geophys. Res., 87, 10,656-10,676, 1982.
Vaughan, T. W., Geologic history of Central America and the West Indies during Cenozoic time, Geol. Soc. Am. Bull., 29, 615-630, 1918.
Wadge, G., and K. Burke, Neogene Caribbean plate rotation and associated Central American tectonic evolution, Tectonics, 2, 633-643, 1983.
Weyl, R., Geology of Central America, 371 pp., Gebruder Borntraeger, Berlin, 1980.

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